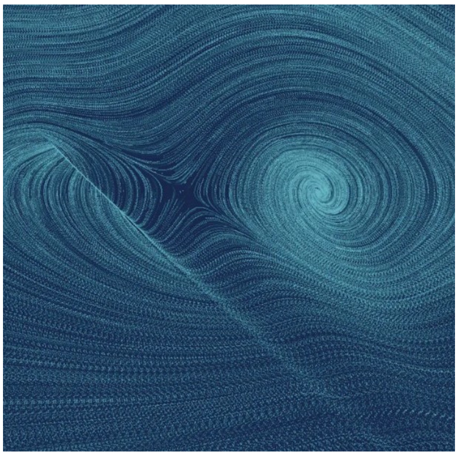

Vector Calculus

The Geometry of Fields and Motion

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1 Curves, Trajectories, and Differential Operators

> ”The dominant idea in vector analysis is that of a vector, and it is the geometric nature of these quantities that makes the subject so powerfully applicable to physics.”
>
> *Vector Analysis*
> J. Willard Gibbs

1.1 Kinematics of Space Curves

Before we can measure how fields flow or circulate, we must understand the paths through which particles travel. By defining curves rigorously, we lay the groundwork for describing motion in three-dimensional space.

1.1.1 Parametrization and Arclength

Definition 1.1 – Arclength of a Space Curve

Let $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$ be a smooth parametrized curve. The physical distance traveled along the curve from $t = a$ to $t = b$ is defined as the integral of the speed:

$$s = \int_a^b \|\mathbf{r}'(t)\|\, dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt \tag{1.1}$$

Remark: Intrinsic Coordinates

As a particle moves, it defines its own intrinsic coordinate system at every point along the curve, known as the Frenet-Serret Frame (Tangent, Normal, and Binormal vectors).

1.2 The Del Operator

We now introduce environments for those curves to exist within: scalar and vector fields.

1.2.1 Divergence and Curl

Theorem 1.1 – The Divergence Theorem

Let V be a simple solid region and let S be the boundary surface of V, given with positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous first partial derivatives. Then the integral of the divergence over the volume equals the flux across the boundary:

$$\iiint_V (\nabla \cdot \mathbf{F})\, dV = \iint_S \mathbf{F} \cdot d\mathbf{S} \tag{1.2}$$

Example 1.1 – Checking the Divergence Theorem

Consider a vector field $\mathbf{F} = \langle x, y, z \rangle$ and let V be the unit sphere $x^2 + y^2 + z^2 \leq 1$. The divergence is $\nabla \cdot \mathbf{F} = 1 + 1 + 1 = 3$. Integrating this over the volume of the sphere gives $3 \times \frac{4}{3}\pi(1)^3 = 4\pi$, perfectly matching the surface flux calculated via the right-hand side.